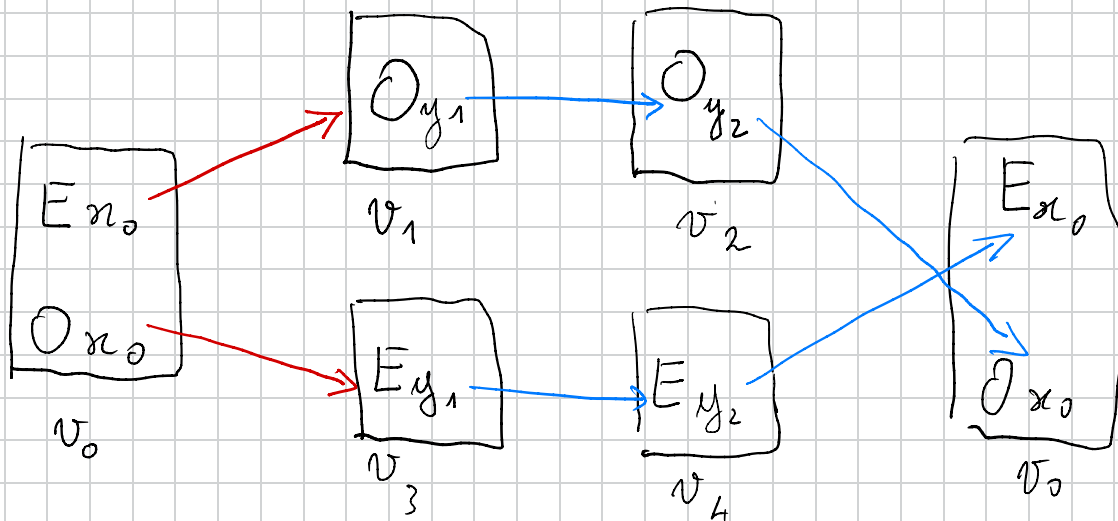
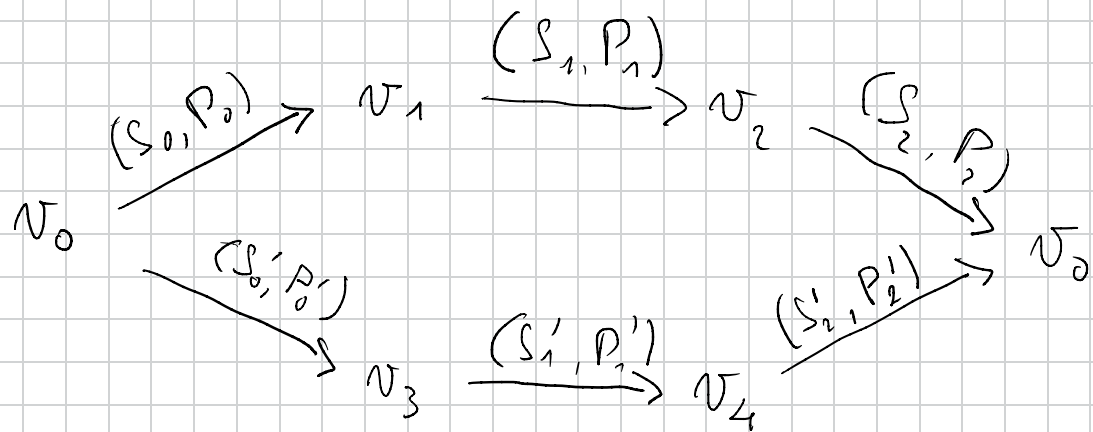
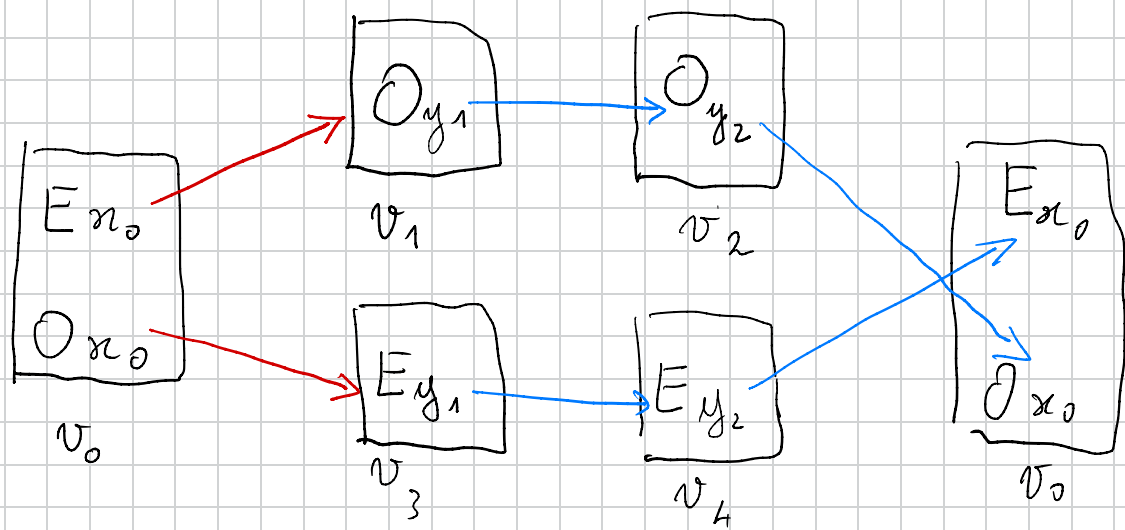


$$\begin{array}{c}
 \frac{}{\vdash N0} (NR_1) \quad \frac{\frac{Ox_o \vdash Nx (\dagger)}{Oy_2 \vdash Ny} (Subst) \quad \frac{Oy_2 \vdash Ny}{Oy_1 \vdash Nsy} (NR_2)}{Oy_1 \vdash Nsy} (Case E) \quad \frac{(*) Ox_o \vdash Nx}{Ey_2 \vdash Ny} (Subst) \quad \frac{Ey_2 \vdash Ny}{Ey_1 \vdash Nsy} (NR_2)}{Ey_1 \vdash Nsy} (Case O) \\
 \frac{Ex_o \vdash Nx (*) \quad (\dagger) Ox_o \vdash Nx}{Ex \vee Ox \vdash Nx} (\vee L)
 \end{array}$$

$$X = \{ \bar{E}x_o, \bar{E}y_1, \bar{E}y_2, Ox_o, Oy_1, Oy_2 \}$$



where Ey_1 is short for $(Ey, 1)$ etc.



$$S_0 = \phi \quad P_0 = \{(E_{x_0}, O_{y_1})\}$$

$$S_1 = \{(O_{y_1}, O_{y_2})\} \quad P_1 = \phi$$

$$S_2 = \{(O_{y_2}, O_{x_0})\} \quad P_2 = \phi$$

$$S'_0 = \phi, \quad P'_0 = \{(O_{x_0}, E_{y_1})\}$$

$$S'_1 = \{(E_{y_1}, E_{y_2})\} \quad P'_1 = \phi$$

$$S'_2 = \{(E_{y_2}, E_{x_0})\} \quad P'_2 = \phi$$

Consider the paths:

$$\pi_1: v_0 \xrightarrow{(S_0, P_0)} v_1 \xrightarrow{(S_1, P_1)} v_2 \xrightarrow{(S_2, P_2)} v_0$$

$$\pi_2: v_0 \xrightarrow{(S'_0, P'_0)} v_3 \xrightarrow{(S'_1, P'_1)} v_4 \xrightarrow{(S'_2, P'_2)} v_0$$

then there are the infinite paths:

$$\pi_1 \pi_1 \dots \pi_2 \pi_2 \dots \text{ and}$$

$$\pi_1 \pi_2 \pi_1 \dots \pi_2 \pi_1 \pi_2 \dots \text{ etc.}$$

that can be described with the generalized regular expression

$$\text{Paths} = (\pi_1 + \pi_2)^{\omega}$$

Let $\pi = \pi_1 \pi_2 \pi_1 \pi_2$ one of these paths; then

the trace

$$z = Ex_0, Oy_1, Oy_2, Ox_0, Ey_1, Ey_2, Ex_0, \dots$$

is compatible with π , $z \models \pi$, and it is infinitely progressing since

$$Ex_0 P_0 Oy_1$$

We also observe that it is compatible with any $\pi' \in \text{Paths}$